

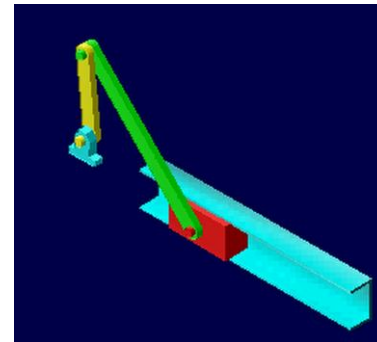
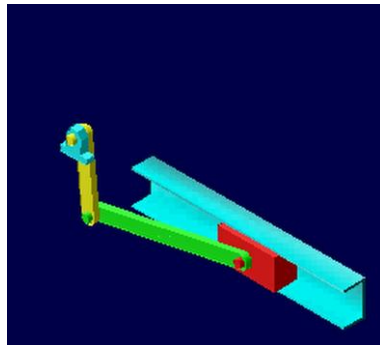
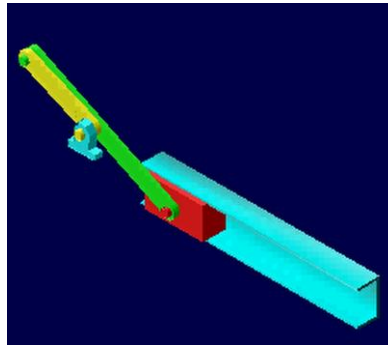
DC Brush Induction Motor Manufacturing – Wire Winding



https://drive.google.com/file/d/10fJcn_xUYaTIWtXCiuJC4MeSeFLULzfj/view

Can you have a slider-crank linkage
in which the slider drives the crank?

How do you control how it comes out of the change point?



What is an everyday example?

https://www.youtube.com/watch?v=BXQ27pU3_7E

With slides from:

- David Remy
- Diann Brei
- Shorya Awtar
- Jeff Stein
- Chinedum Okwudire

Actuators and Motors

Design + Manufacturing II
ME 350 Lectures #6 and #7

Example of a MATLAB program to solve for one dyad

% Dyad Synthesis -- Boat Example

% Assign a variable to convert degrees to radians

deg = pi / 180;

% define i

i = sqrt(-1);

Convert to radians
(MATLAB works in radians)



% Free choices (change in orientation of links attached to ground)

beta2 = 50 * deg;

beta3 = 200 * deg;

Randomly chosen values
of β_2 and β_3 (*Free choices*)



% Knowns (change in orientation and position of coupler link)

alpha2 = 11 * deg;

alpha3 = 0 * deg;

delta2 = -58 + 25*i;

delta3 = -89 - 26*i;

% Calculations (using Cramer's rule to solve matrix equations)

A = [delta2, exp(i*alpha2)-1; delta3, exp(i*alpha3)-1];

B = [exp(i*beta2)-1, delta2; exp(i*beta3)-1, delta3];

C = [exp(i*beta2)-1, exp(i*alpha2)-1; exp(i*beta3)-1, exp(i*alpha3)-1];

% And the final answer is... (don't forget to repeat for the other dyad, by simply choosing different betas)

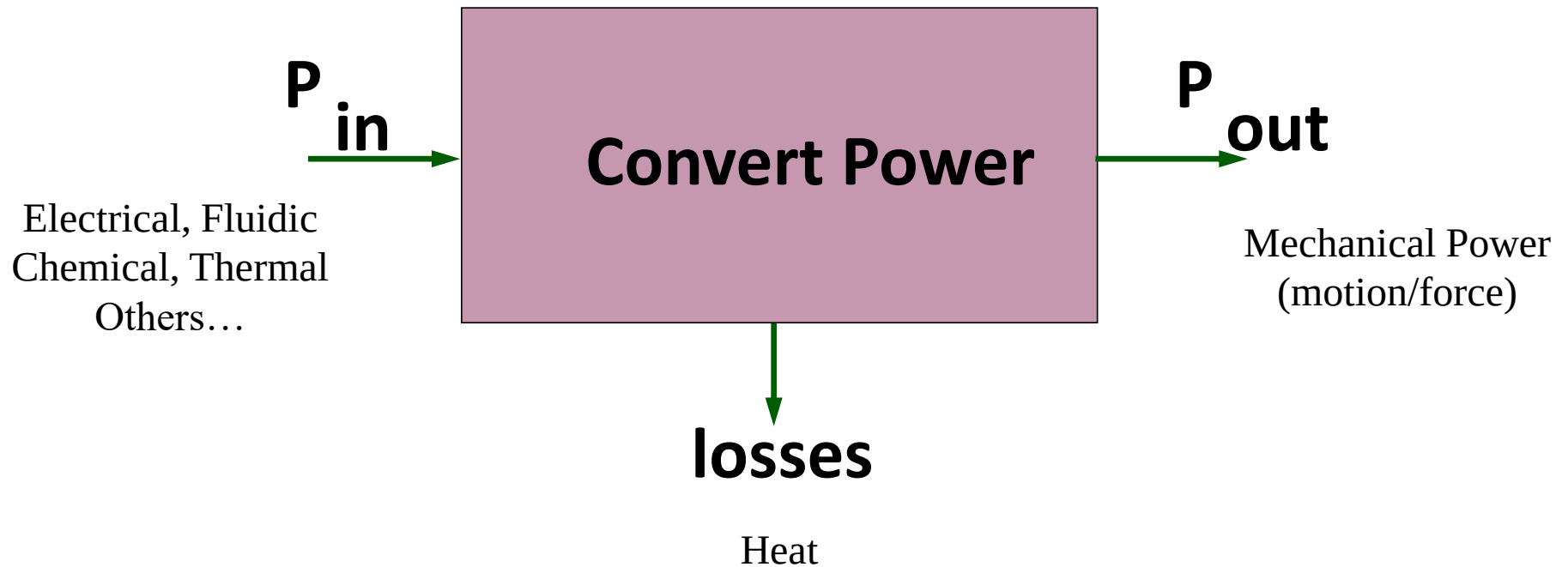
W = det(A) / det(C);

Z = det(B) / det(C);

Goals for this lecture and the next one:

- Explore common types of actuators
- Deep dive into DC motors
 - Derive the torque-speed relationship
 - Calculate how fast a DC motor will turn depending on voltage and load
 - Quantify the torque-current relationship
 - Quantify back-EMF
 - Quantify losses and efficiency
 - Identify physical performance limits of motors

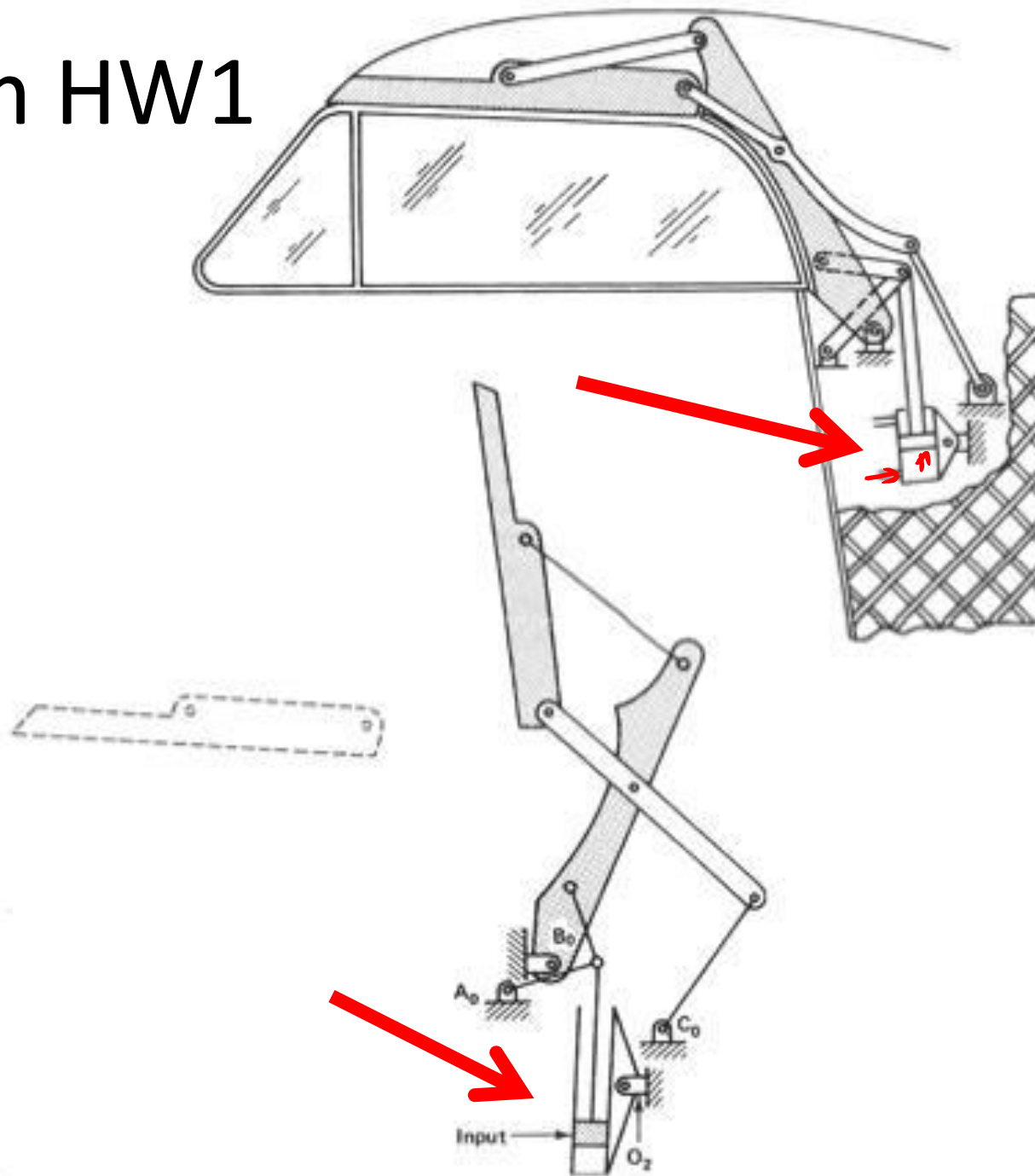
Actuator: Functional Block



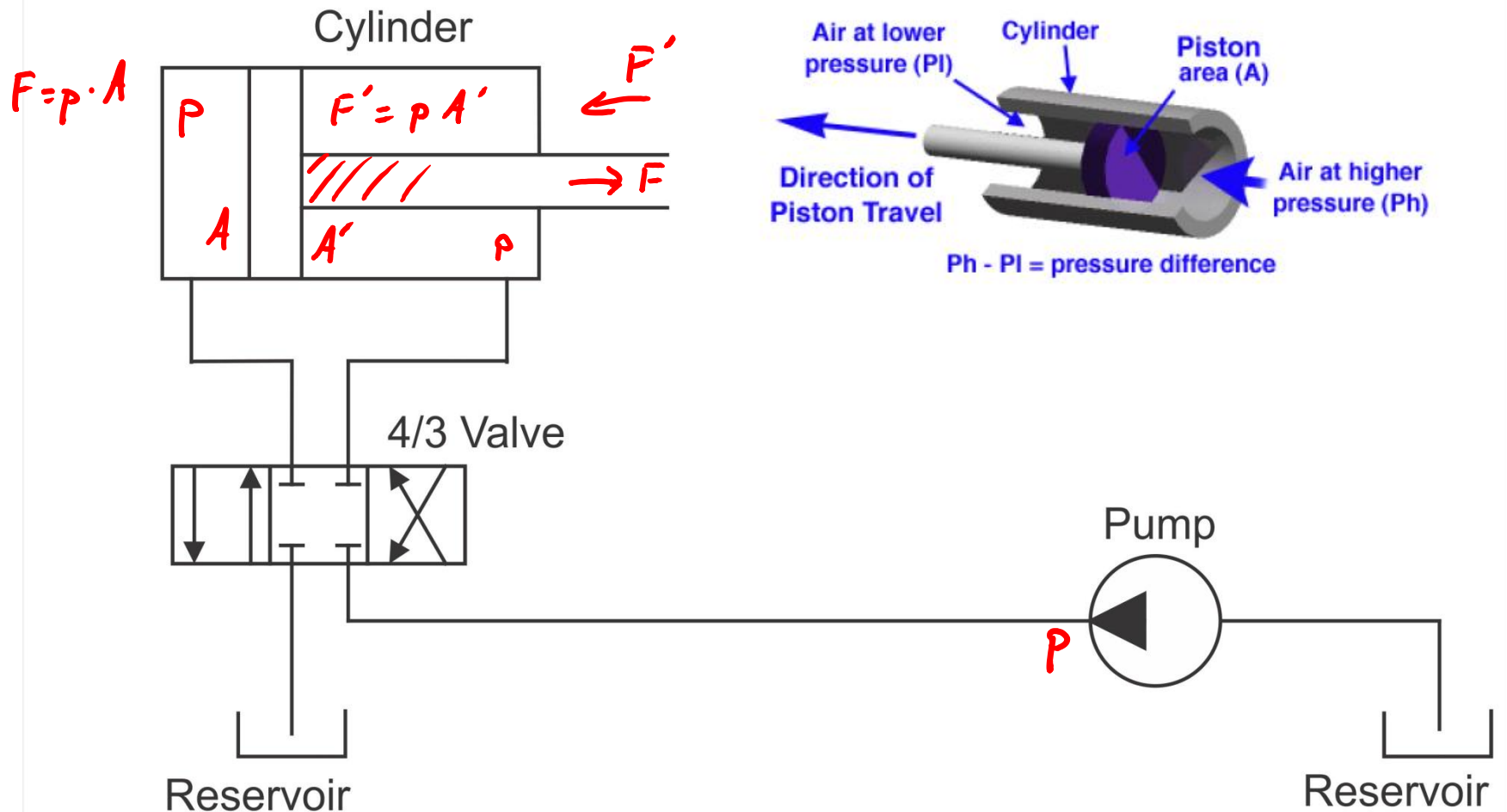
Hydraulic Actuators (Cylinders)



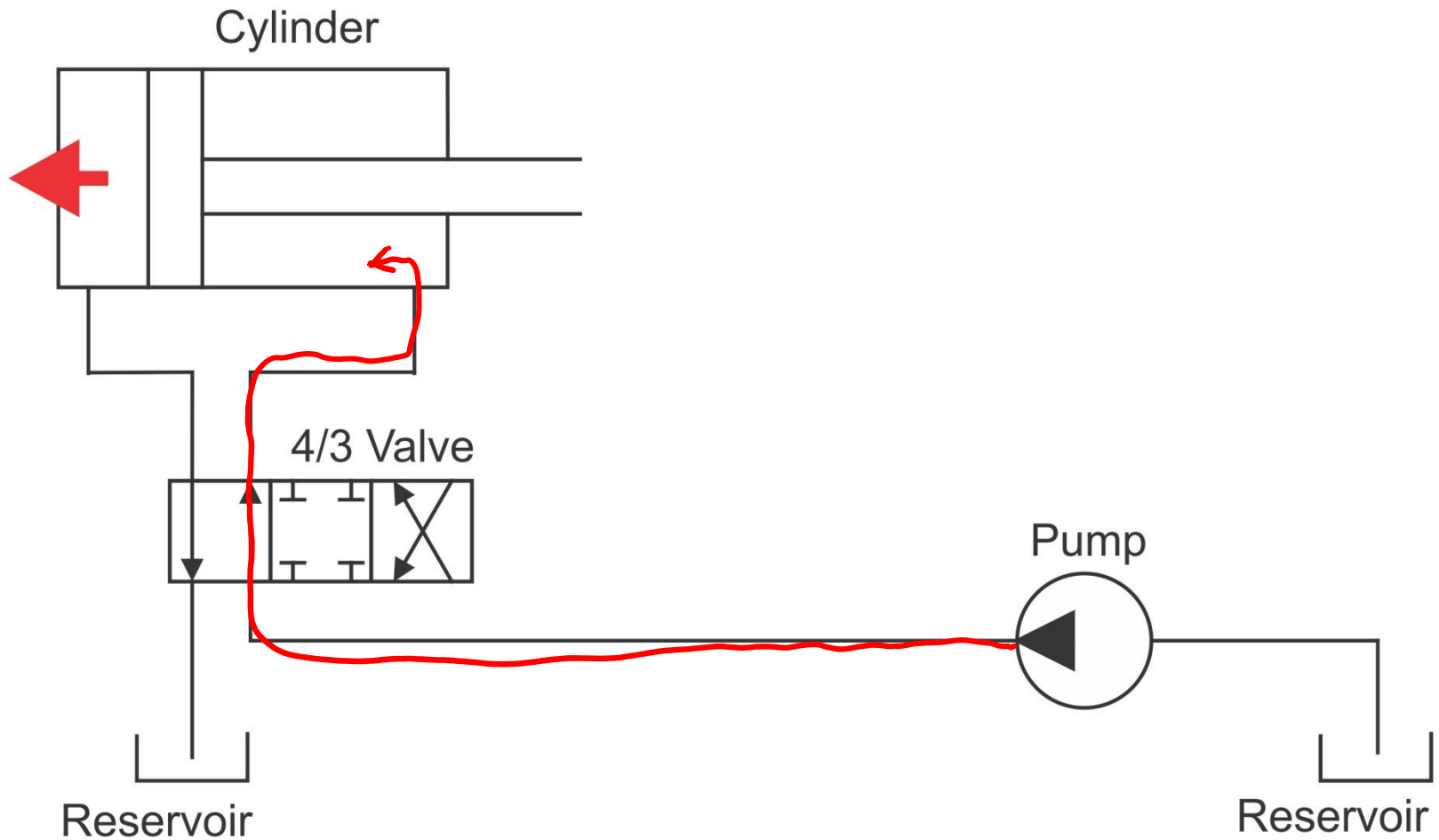
From HW1



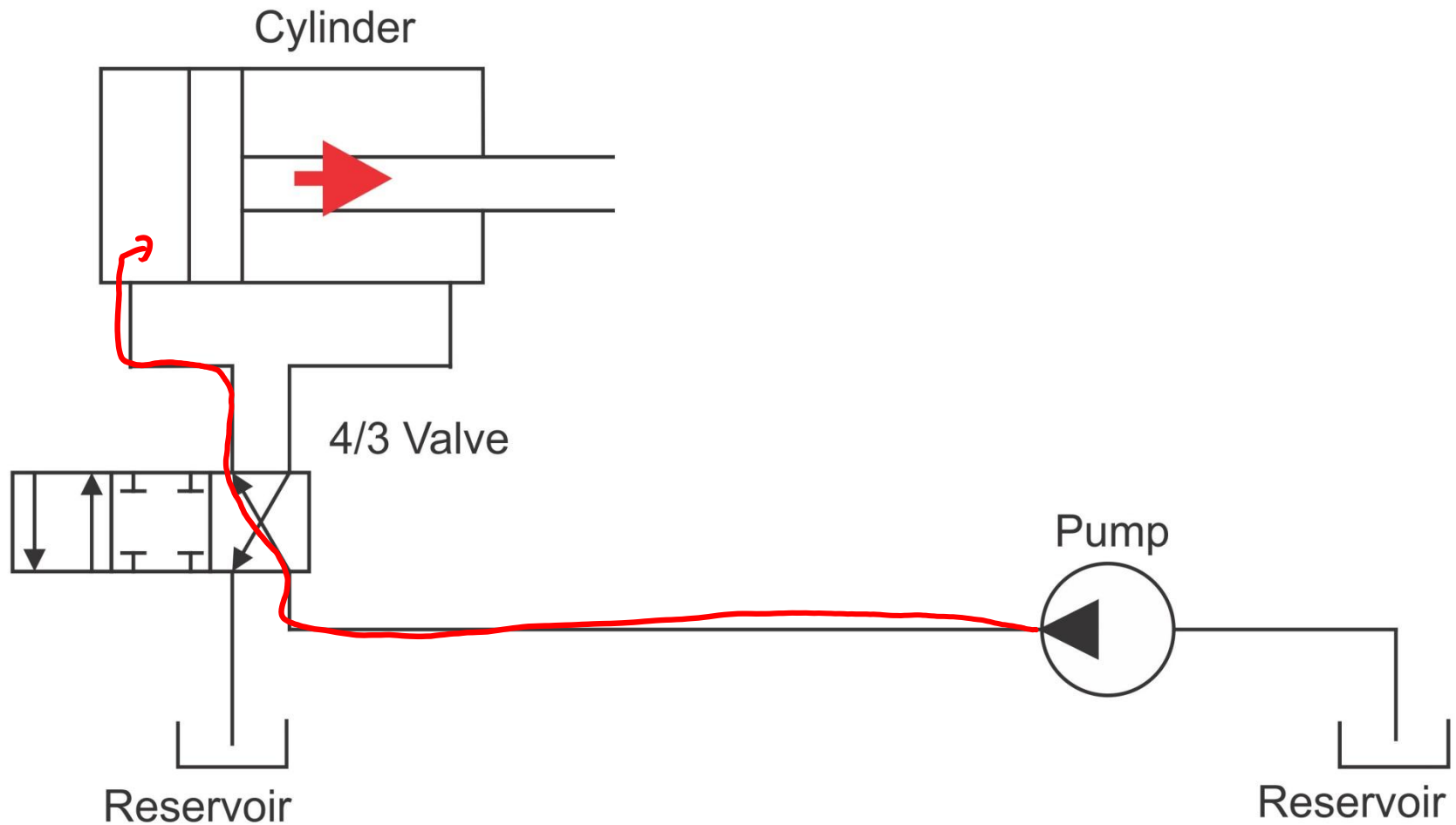
Hydraulic components



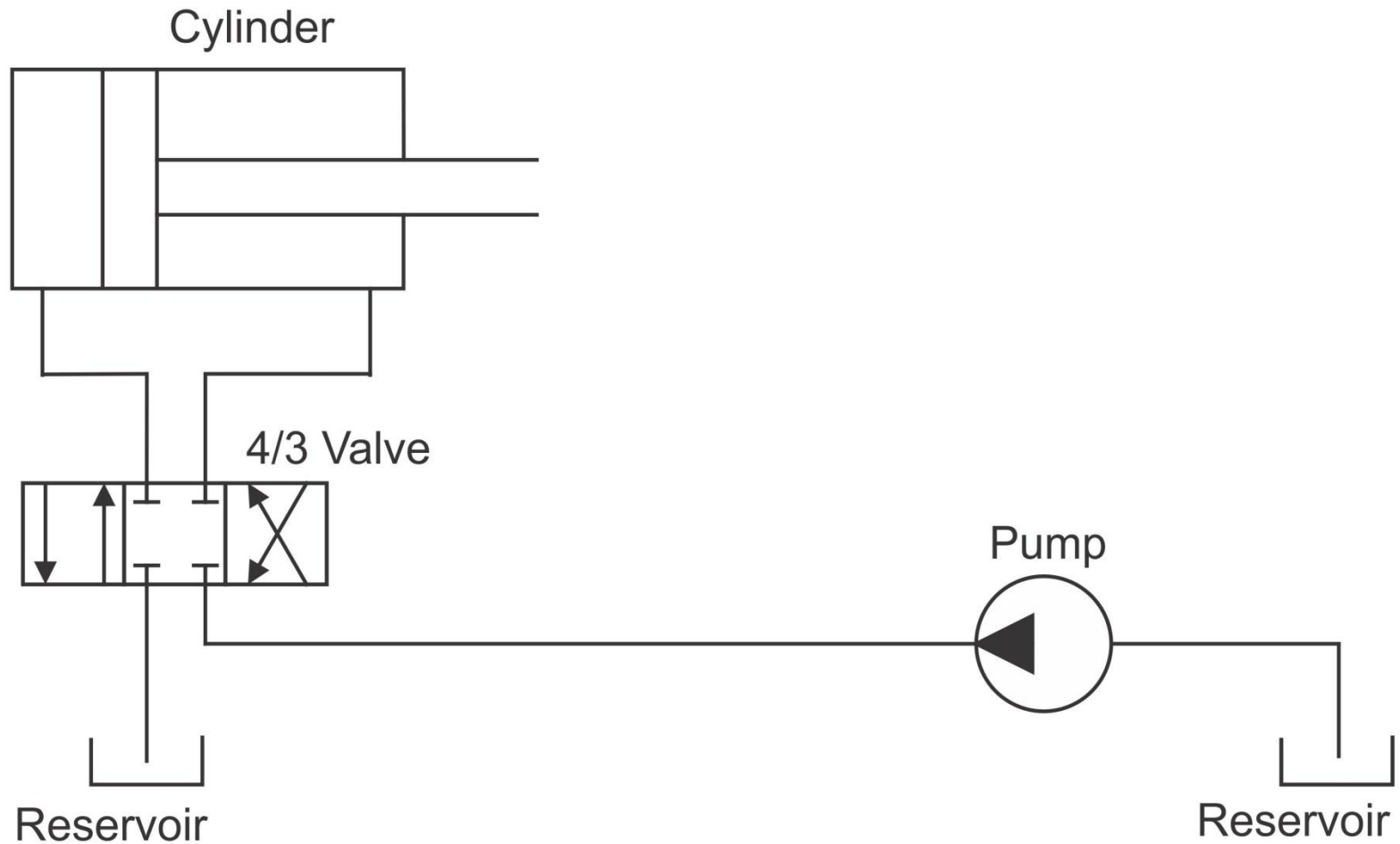
Hydraulic components



Hydraulic components



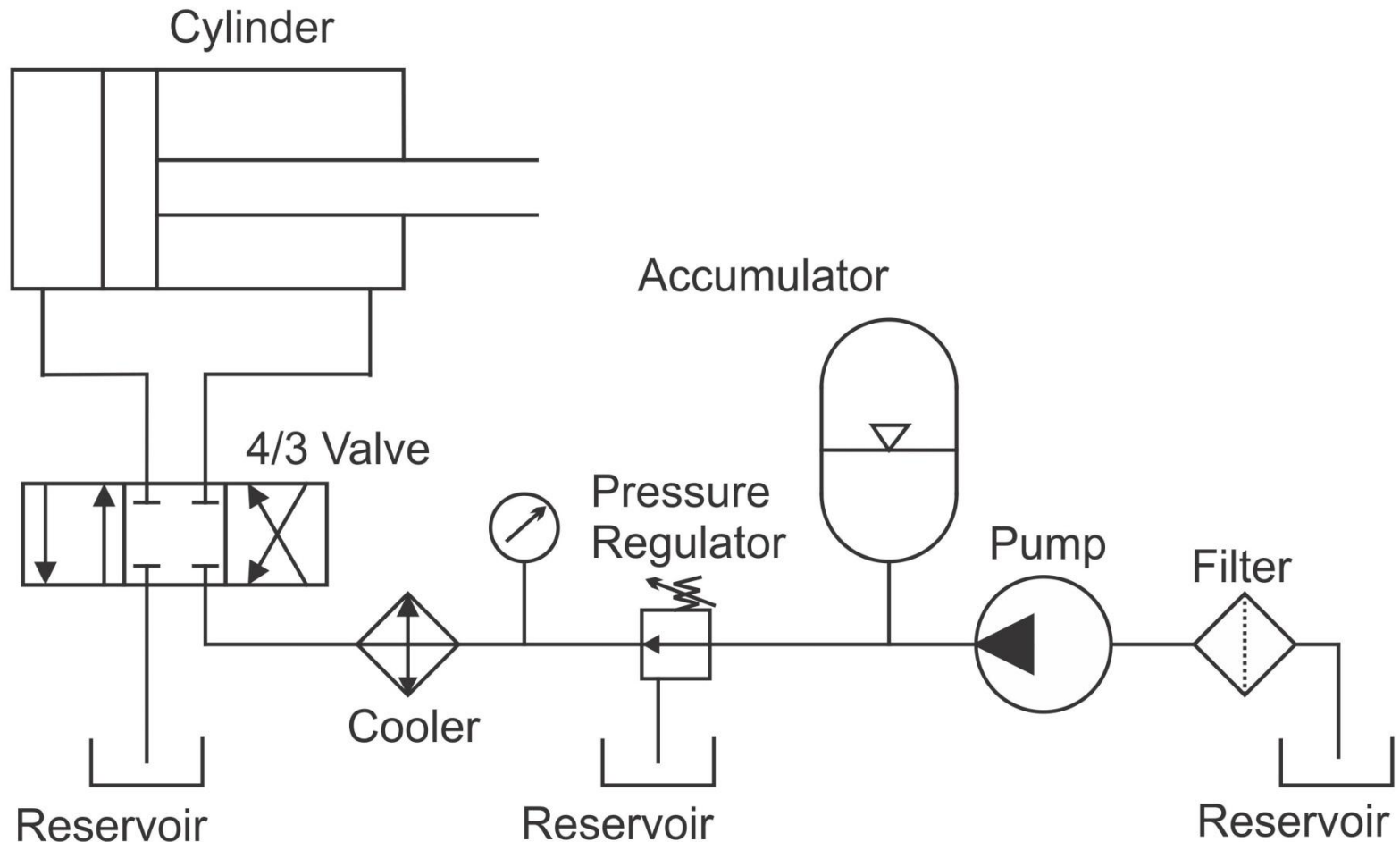
Hydraulic components



Hydraulic components

Pressure Regulators adjust the force exerted by the piston.

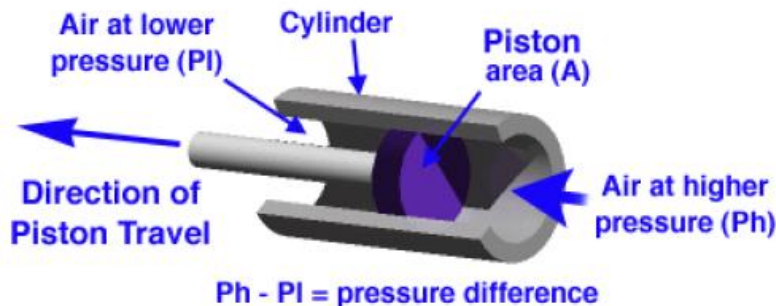
Flow Control Valves adjust the speed of the piston.



Hydraulic and Pneumatic

Pneumatic and Hydraulic actuators rely on a pressure differential in two portions of the device to create motion. Typically we use valves to control the pressure in a line from the pump or source.

- **Energy source can be remote to a light weight actuator** but the pump/source infrastructure is needed, which can be bulky and loud.
- Typically linear but ultrasonic and rotary architectures are possible (been to the dentist lately? (high speed air drills))
- Very large forces and strokes are possible, speed is moderate, good for harsh environments
- Can be modeled or characterized - is mature technology
- Issues with sealing, leakage, “stick”, long term lifetime, extra maintenance and overhead
- Pneumatic cylinders are highly inefficient
- Hydraulics are coming back in fashion. Very useful in Robotics

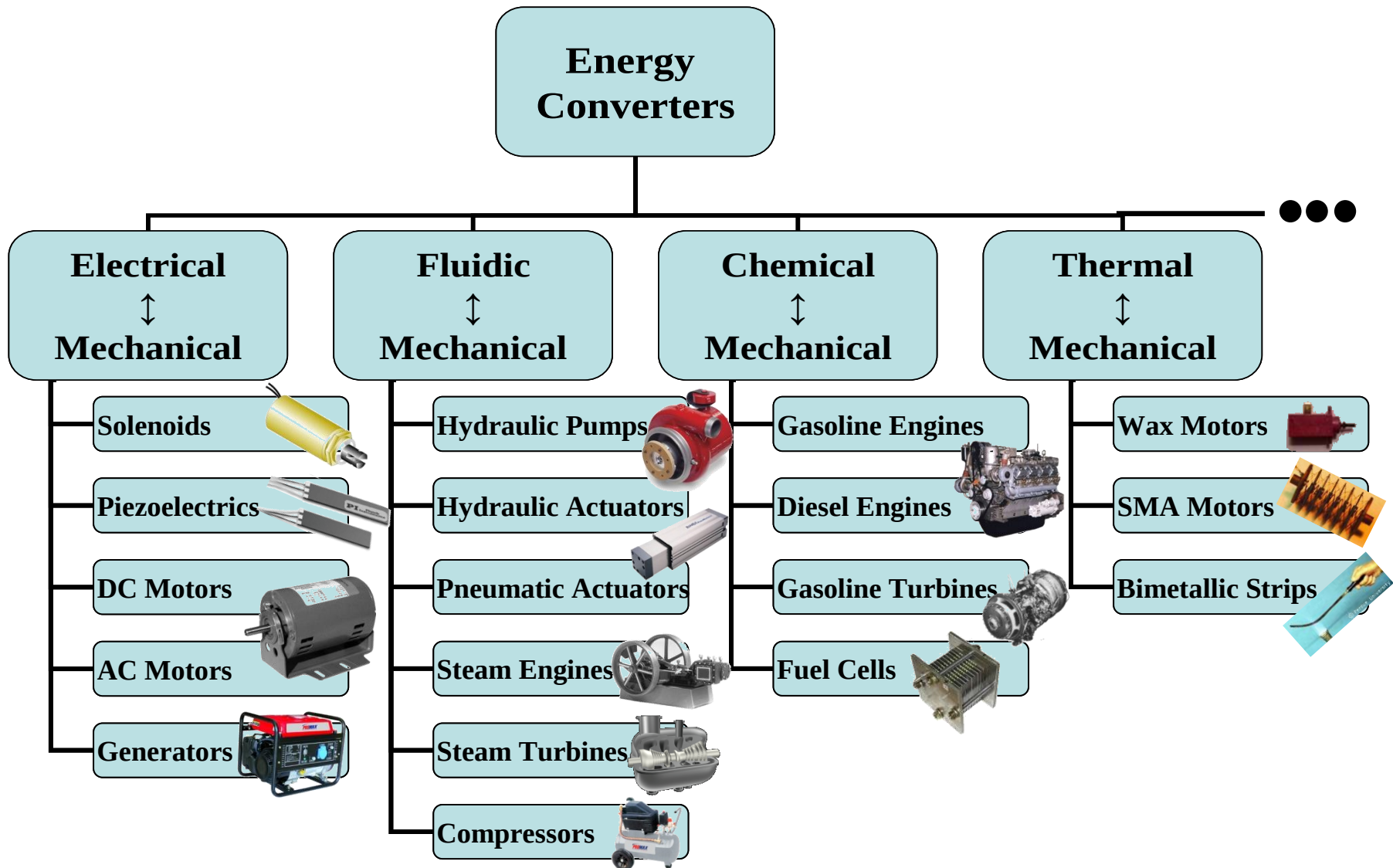


The Hydraulic Power Unit can serve as a counterweight.



Source: <http://www.youtube.com/watch?v=DTw3vpbFJOc>

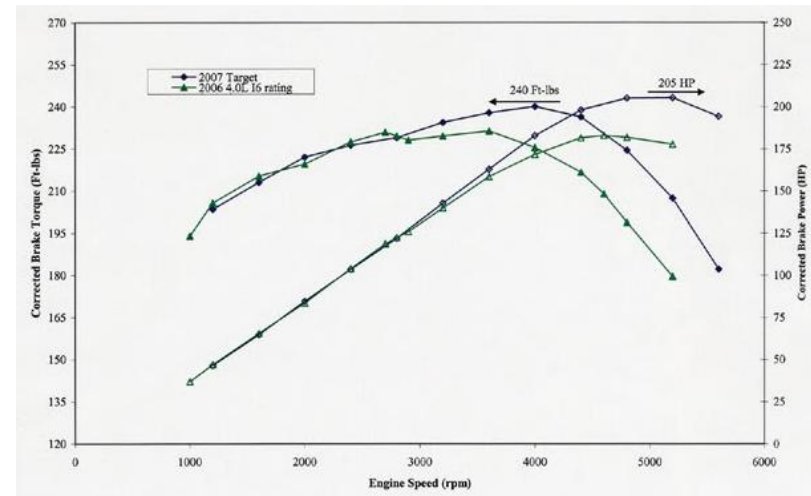
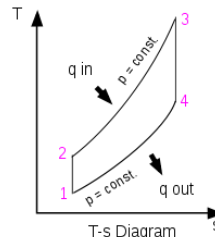
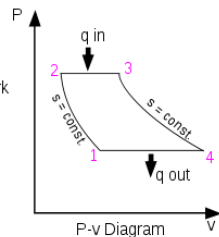
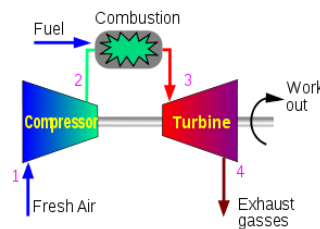
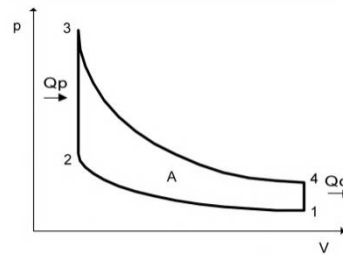
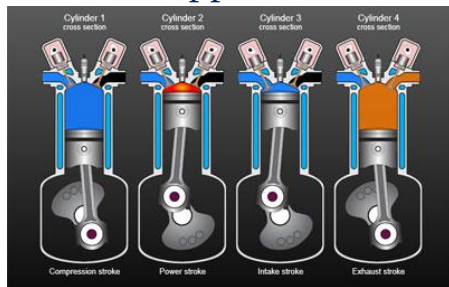
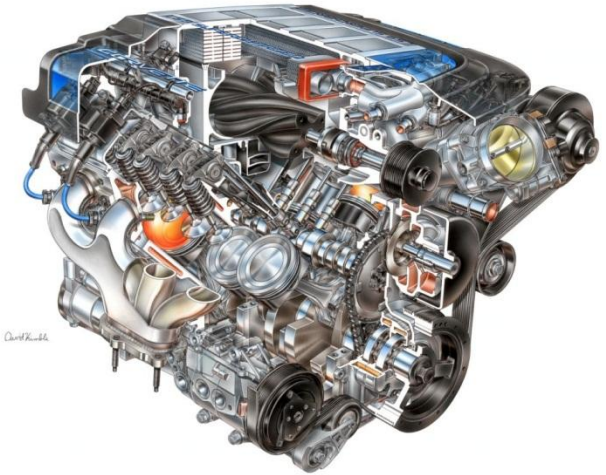
Power/Energy Converters



Chemical Combustion

The internal combustion engine is an engine in which the combustion of a fuel (normally a fossil fuel) occurs with an oxidizer (usually air) in a combustion chamber.

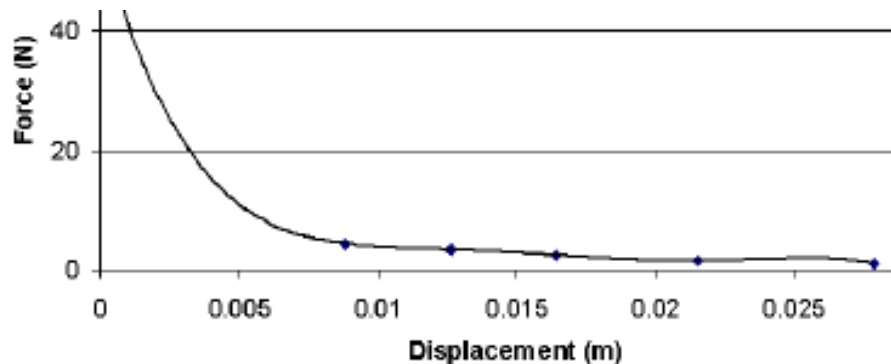
- The expansion of the high-temperature and -pressure gases produced by combustion applies direct force to some component of the engine, such as pistons, turbine blades, or a nozzle.
- It can provide **high power-to-weight ratios** together with **excellent fuel energy density**.
- The real strength of internal combustion engines is thus in mobile applications .



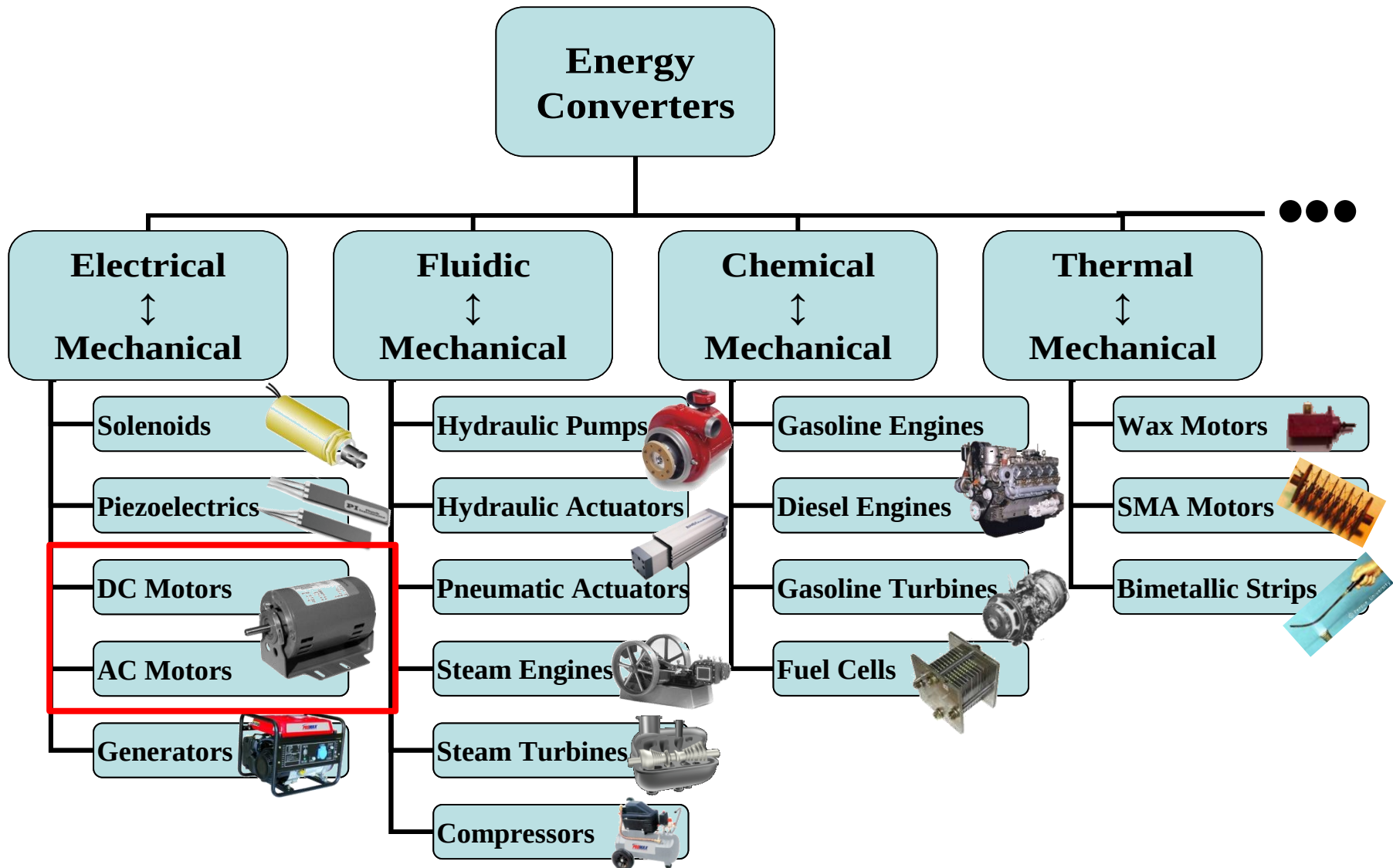
Solenoids

Solenoids use a coil to generate a magnetic field that then attracts an iron plunger to create linear motion.

- They have limited pull force and stroke and are often used to open or close devices or clutches
- The force drops off rapidly (nonlinearly) with motion of the plunger
- Like other electrical motors – they can produce medium forces, strokes and speeds
- The force/displacement curves can be predicted, but typically just use curves from the manufacturer
- A voice coil works with a similar principle, with the coil being the moving part



Power/Energy Converters

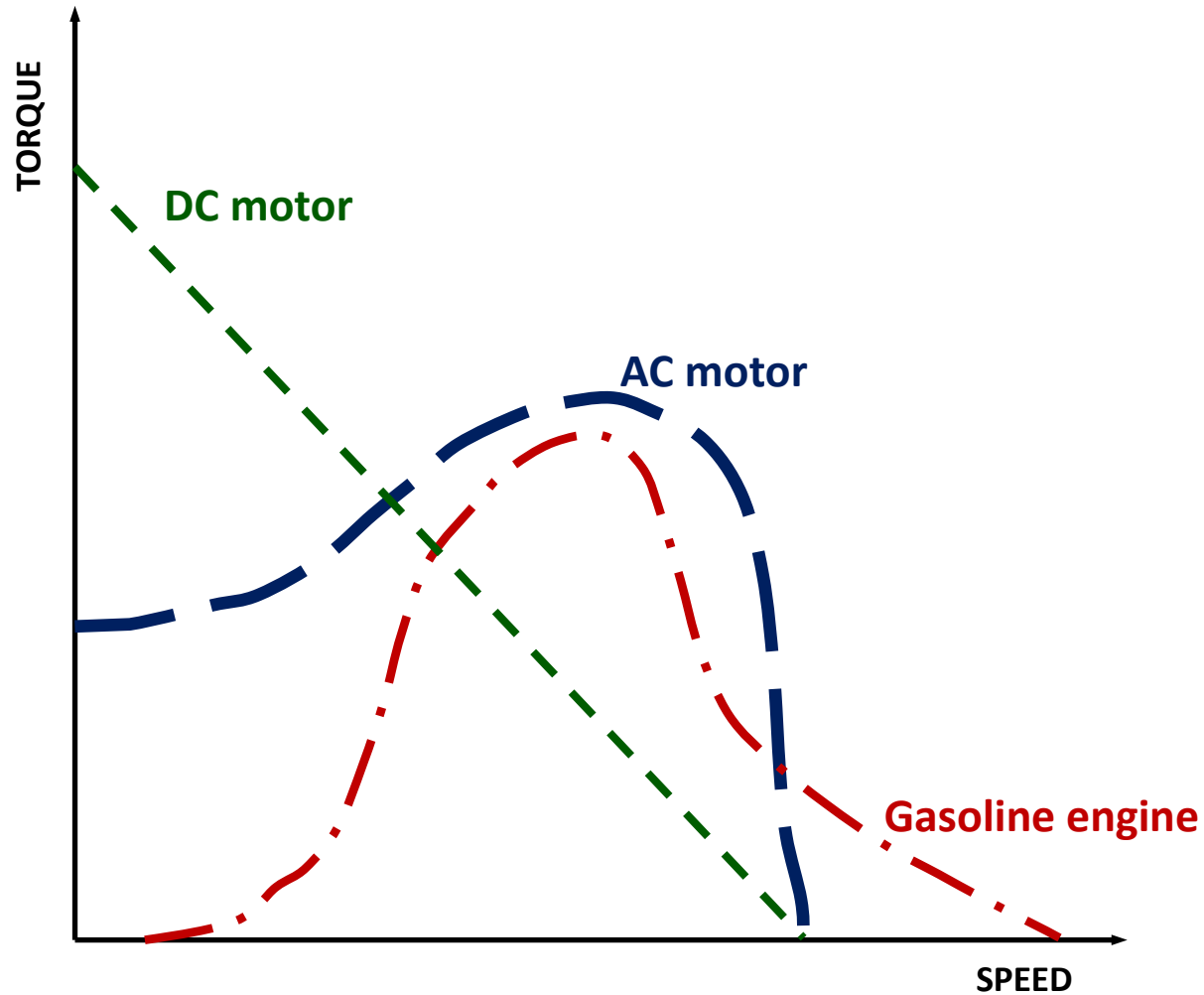


Why Electric Motors?

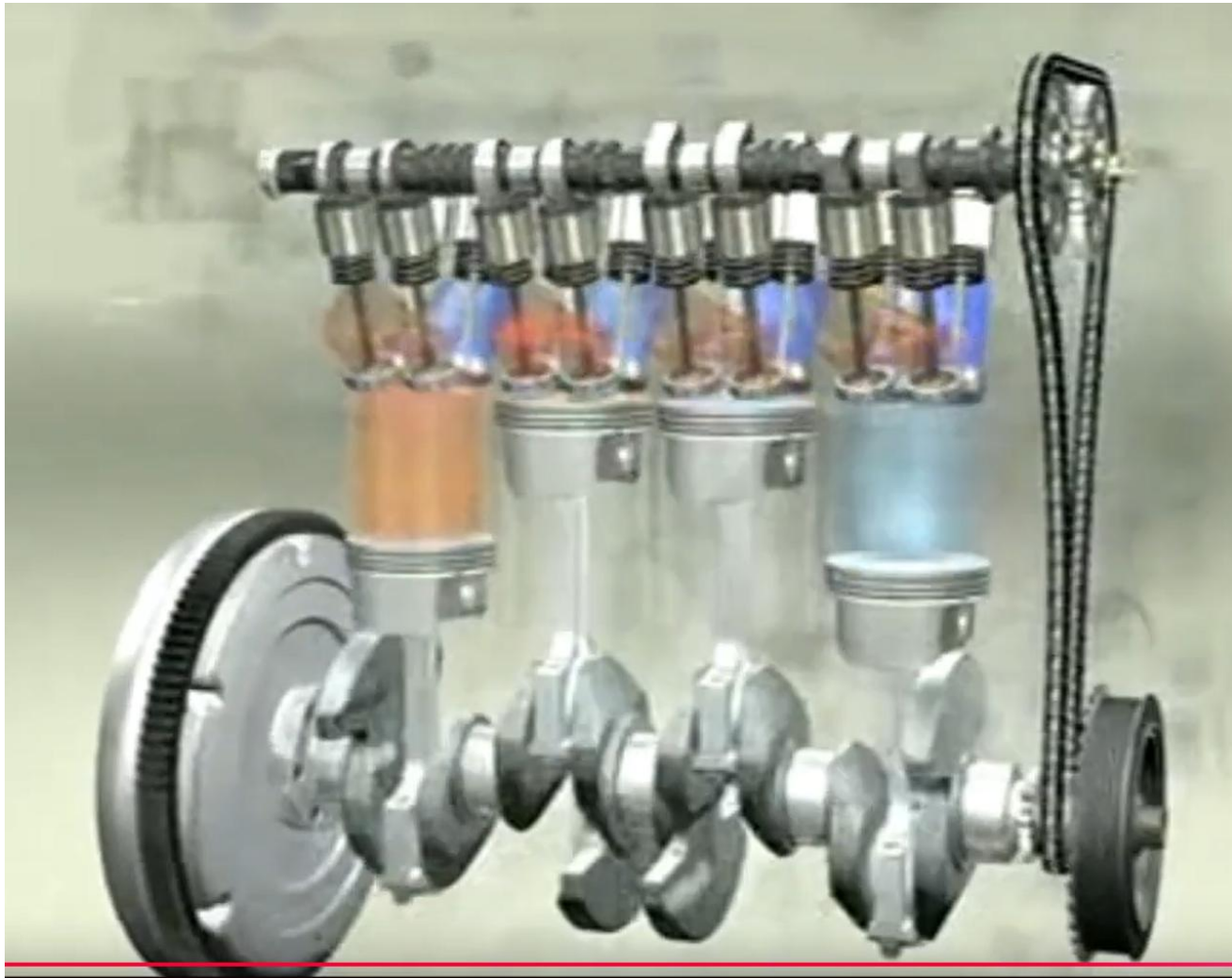
- How many combustion engines are in a car?
- How many electric motors are in a car?



Torque-Speed Curves of Motors



The Flywheel or Flexplate has gear teeth that engage with the teeth of the starter motor.



STRETCH!



Multi
phase

Electrical Motor Categories

Single
phase

~ 1 kW

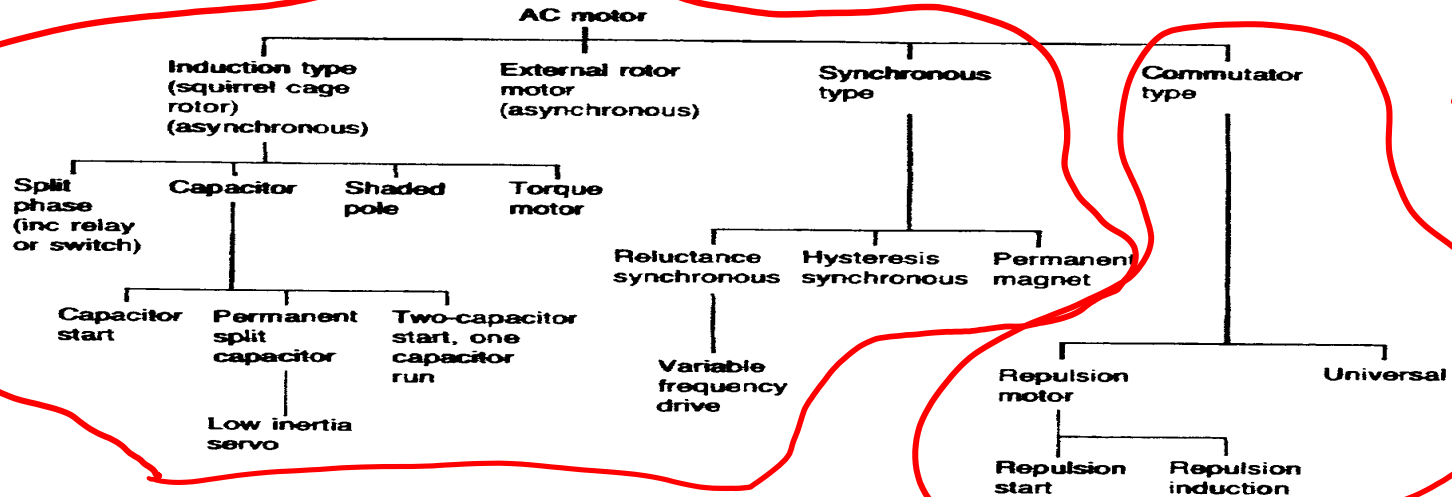
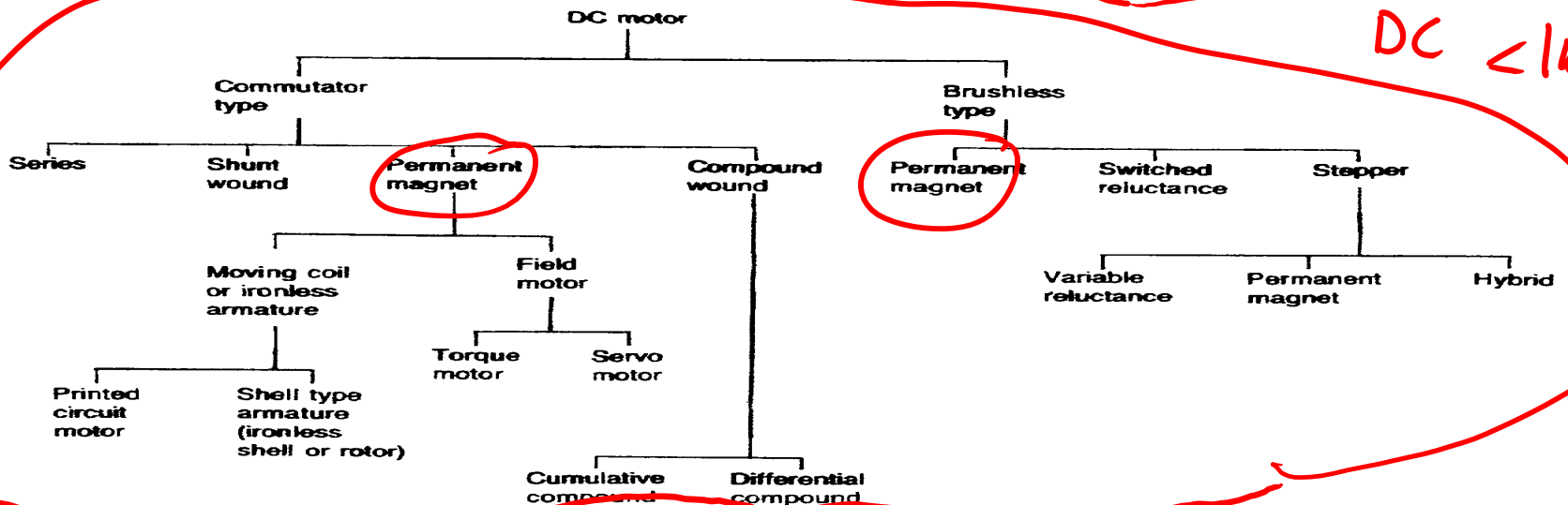
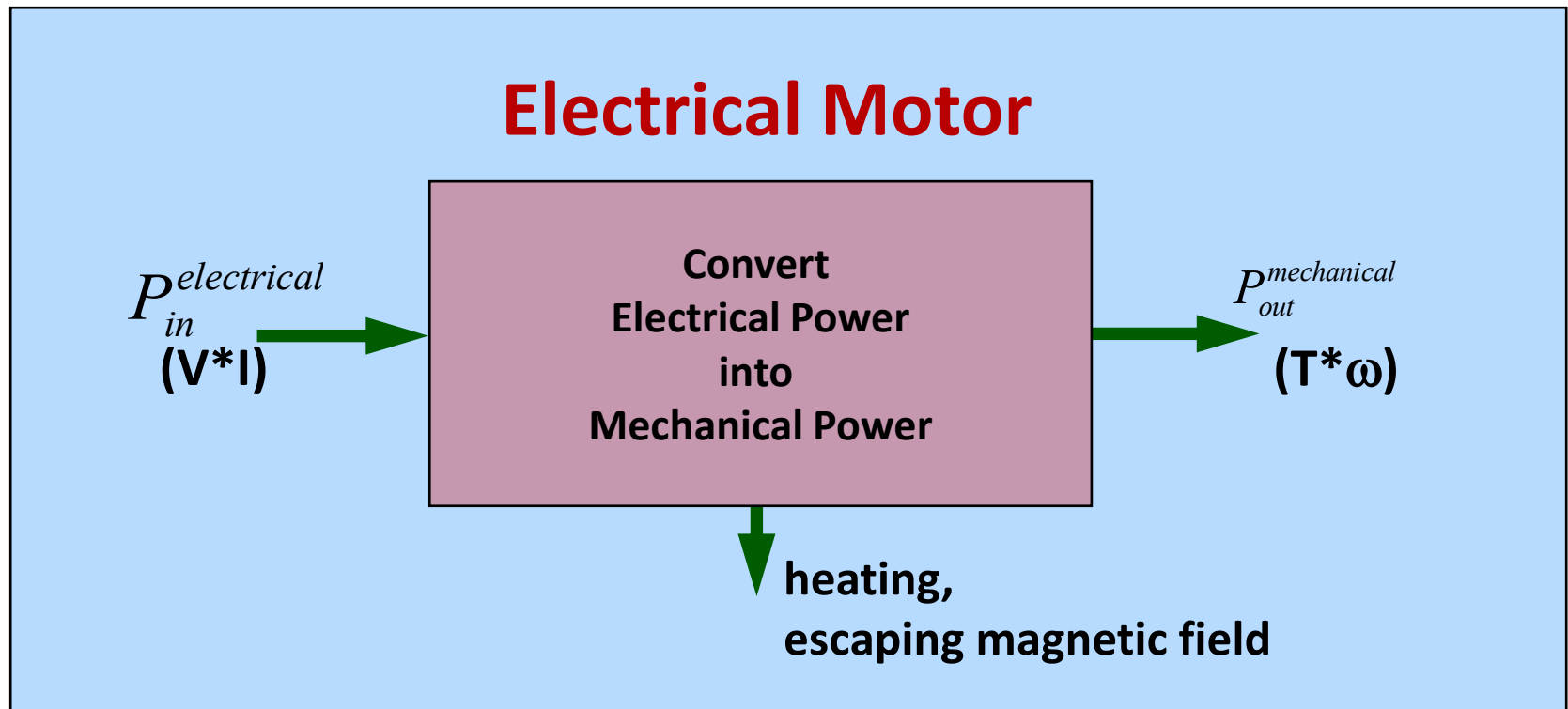
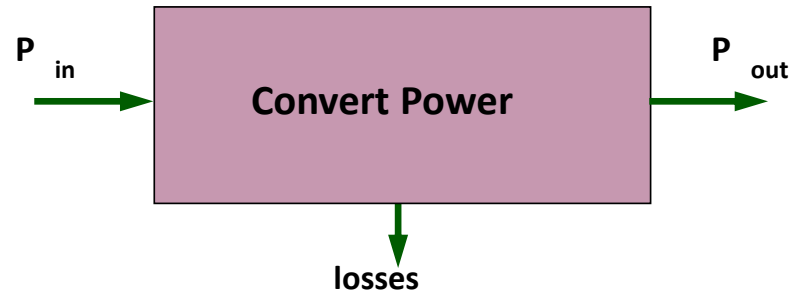


Fig. 8.1 Family tree of a.c. motors

DC < 1 kW



Electrical Motor: Functional Block

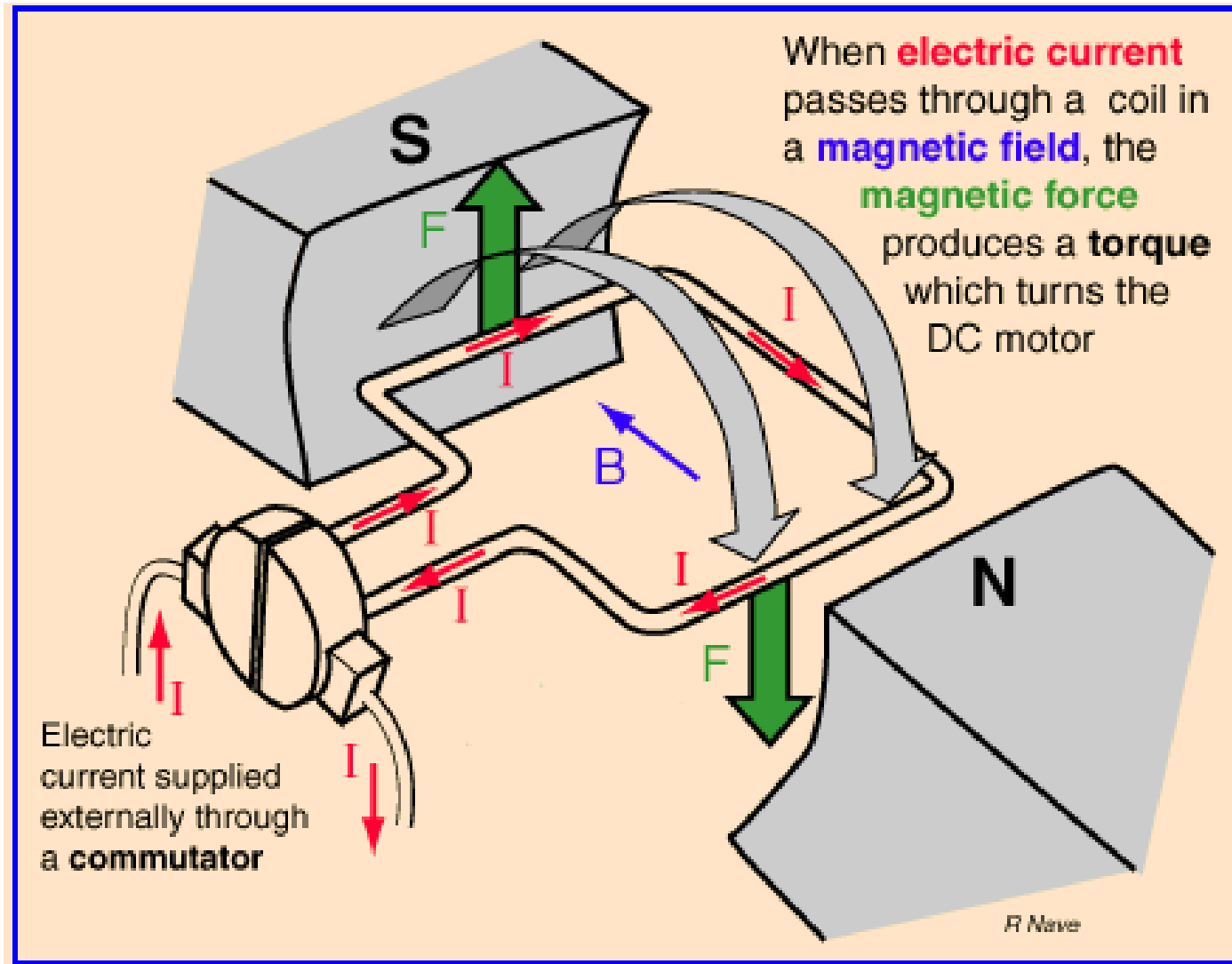


DC Brush Induction Motor - Principle



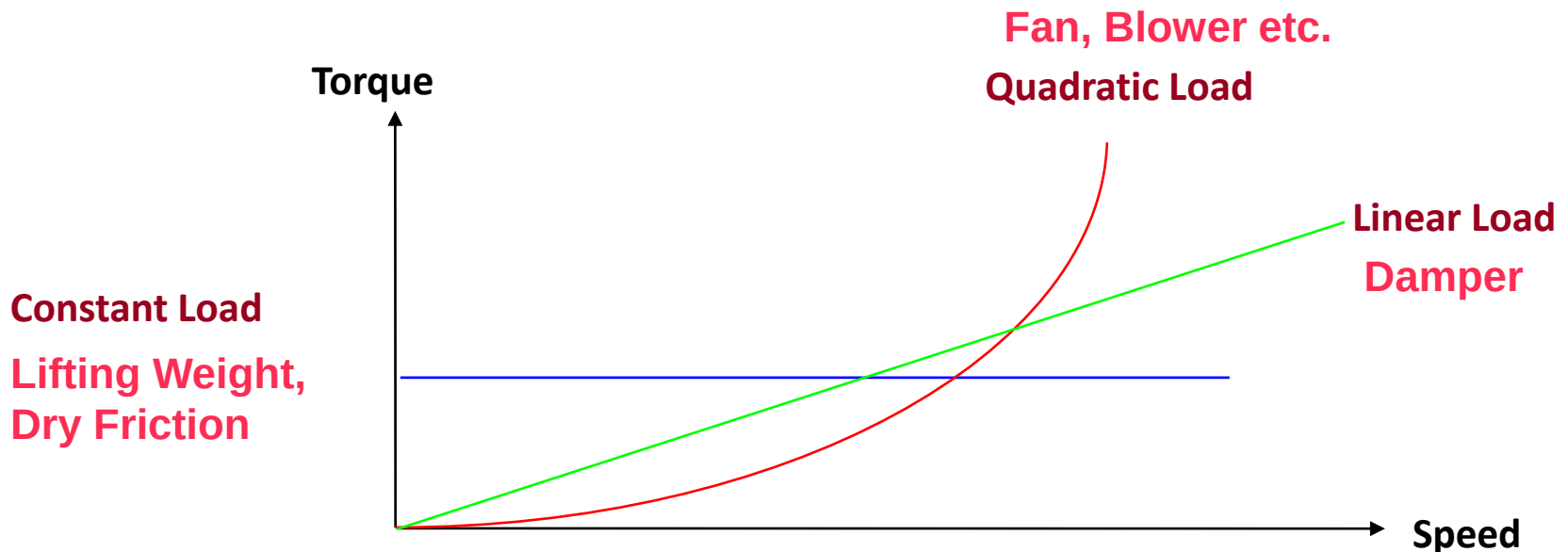
<https://www.youtube.com/watch?v=LAAtPHANefQo>

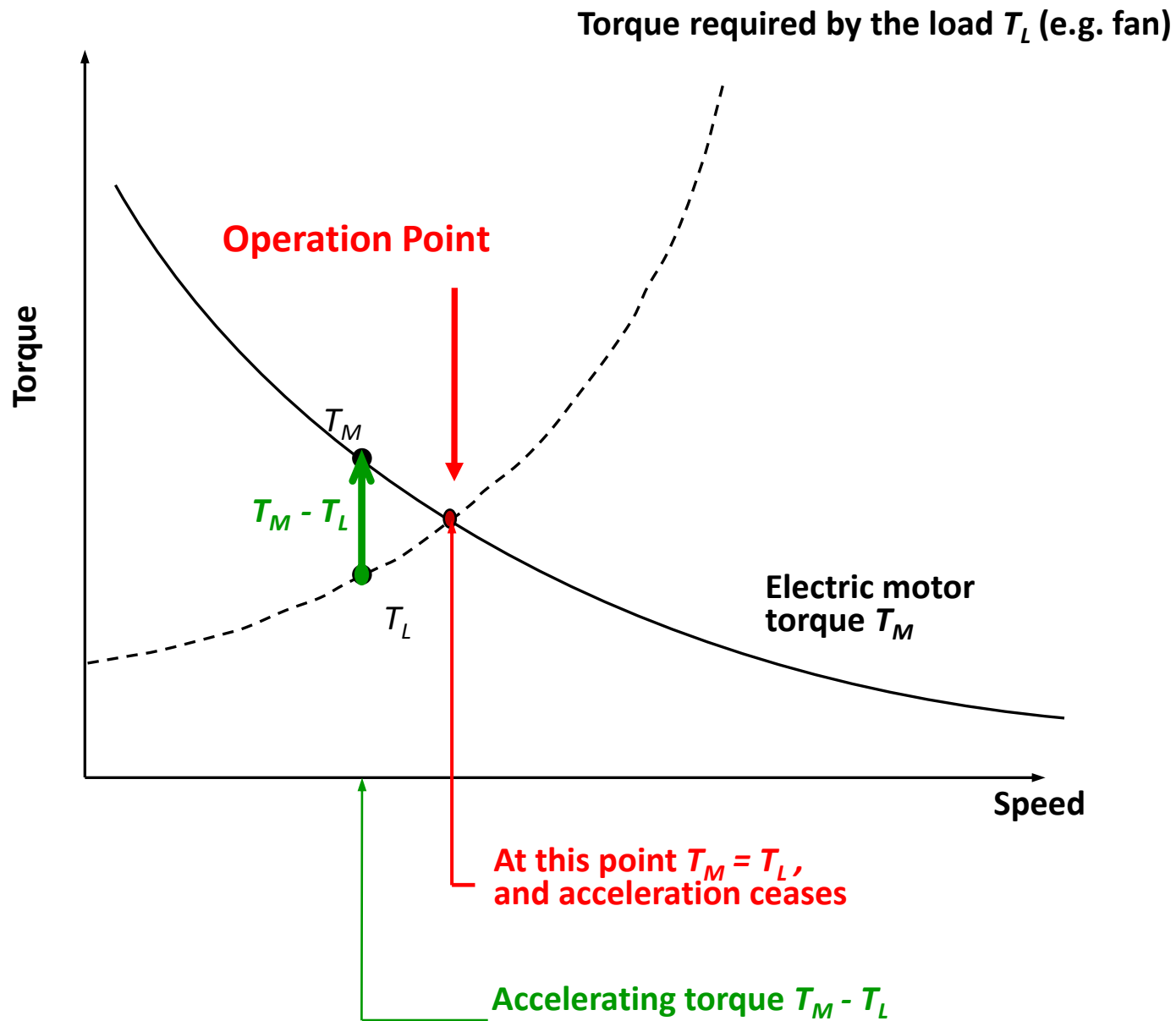
Permanent Magnet DC-Motor



External System Load Lines

- **Constant** - Constant loads appear when the torque (or force) is independent of speed. An example is when a weight has to be lifted or a load moved against dry “Coulomb” friction.
- **Linear** - Linear loads appear when the load torque is proportional to the speed. An example is moving a load against viscous or wet friction (laminar flow). E.g.; stirring pancake batter with an egg-beater or a rotating shaft supported by bearings.
- **Quadratic** - Quadratic loads occur when the load torque is proportional to the square of the speed and are often a result of turbulent flows: examples would include fans, the air drag on aircraft and automobiles.





Using Torque-Speed Curves

The motor operates where the load curve and the torque-speed curve intersect.

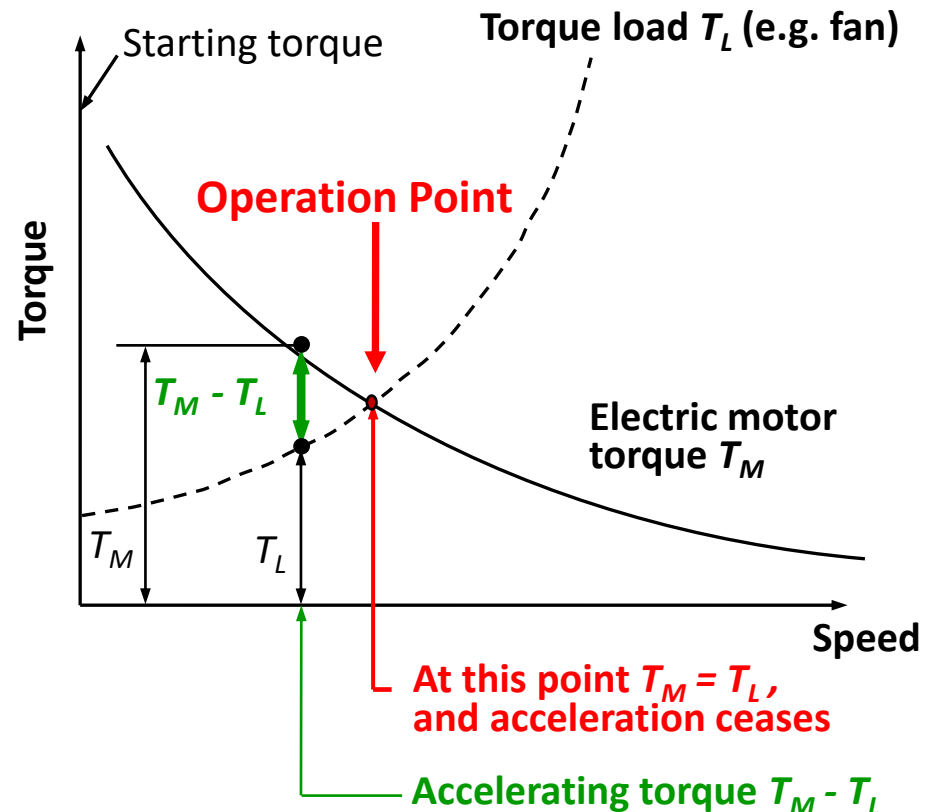
Two main operations regions:

I. Start Up Torque

This is the torque required to change a speed in a motor. Normally used for starting a motor from a dead stop. Inertia and response time is important.

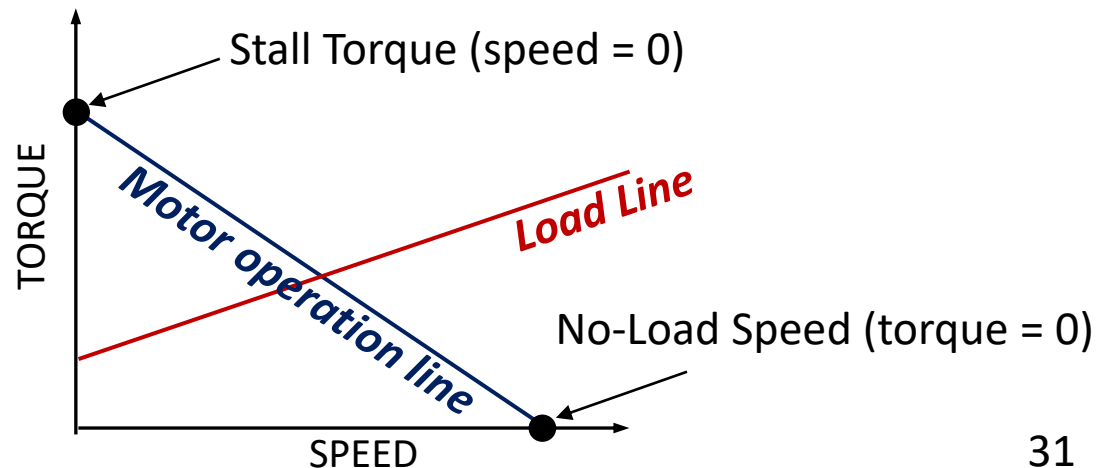
II. Operating Torque

This is the torque required to operate the motor at a constant speed. Inertial loads are zero at constant speeds.



Torque-Speed Curve Terminology

- **Stall Torque** - The maximum torque that the motor can deliver occurs when the motor is loaded so much it comes to a stop.
- **No-Load Speed** - The speed the motor would attain if it were allowed to run with absolutely no external load on it.
- **Speed Regulation** – describes how well a motor can maintain its speed when the amount of load changes.
 - If a motor has good Speed Regulation, is the slope of its operation line steep or shallow?



DC Motor: Back-EMF

- Torque produced by a Permanent Magnet DC motor is directly proportional to the current flowing through its rotor coil
- When a conductor moves in a Magnetic Field, an EMF is generated across it (Faraday's Law of Induction) that tends to oppose the cause that produces the conductor motion (Lenz's Law)
 - This is the same as the generator / tachometer / galvanometer effect
 - The induced EMF is known as the Back EMF in the context of DC motors
- For a permanent magnet DC motor, with commutation, and a multiple coil rotor, the back EMF may be shown to be proportional to Motor Speed

Torque Constant K_t and Back-EMF Constant K_b

- Force generated on a wire and therefore torque generated on the rotor winding is proportionally dependent on the current flowing through it

$$T_m = K_t \cdot i_m$$

- Back emf generated on the rotor winding is dependent on angular speed at which it turns

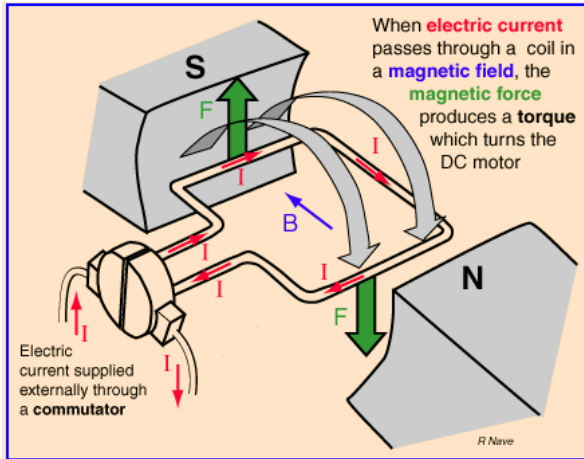
$$V_b = K_b \cdot \omega$$

- Torque constant and back emf constant are equal in SI units. These constants depends on the motor size and design parameters. (number of coils, size of rotor winding, strength of permanent magnet)

$$K_t = K_b$$

Steady-state Operating Equation for PMDC Motor

Equation will be derived in class



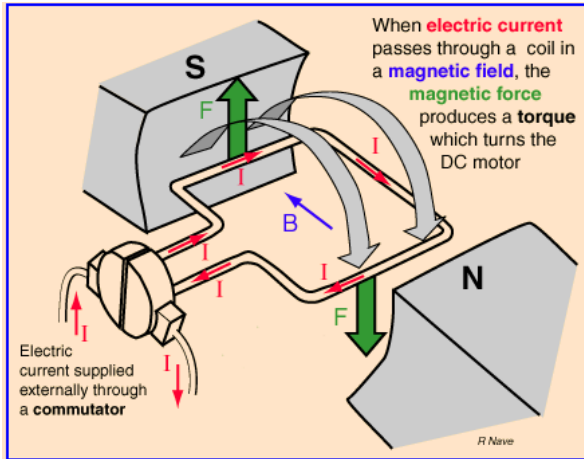
- T_m : Motor Output Torque
- ω : Motor Output Speed
- V_s : Supply voltage
- i_m : Motor Winding Current
- R : Winding Resistance
- L : Winding Inductance
- V_b : Back EMF
- V_R : Voltage across resistor
- V_L : Voltage across inductor
- K_t : Motor's torque constant
- K_b : Motor's back EMF constant
- T_s : Stall torque
- K : Torque-speed gradient

Note that $K_t = K_b$ (if expressed in N-m/A and V/(rad/s), respectively)



Steady-state Operating Equation for PMDC Motor

Equation will be derived in class



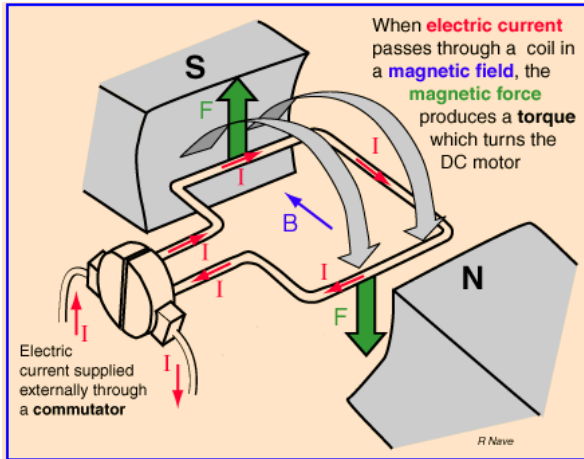
- T_m : Motor Output Torque
- ω : Motor Output Speed
- V_s : Supply voltage
- i_m : Motor Winding Current
- R : Winding Resistance
- L : Winding Inductance
- V_b : Back EMF
- V_R : Voltage across resistor
- V_L : Voltage across inductor
- K_t : Motor's torque constant
- K_b : Motor's back EMF constant
- T_s : Stall torque
- K : Torque-speed gradient

Note that $K_t = K_b$ (if expressed in N-m/A and V/(rad/s), respectively)



Steady-state Operating Equation for PMDC Motor

Equation will be derived in class

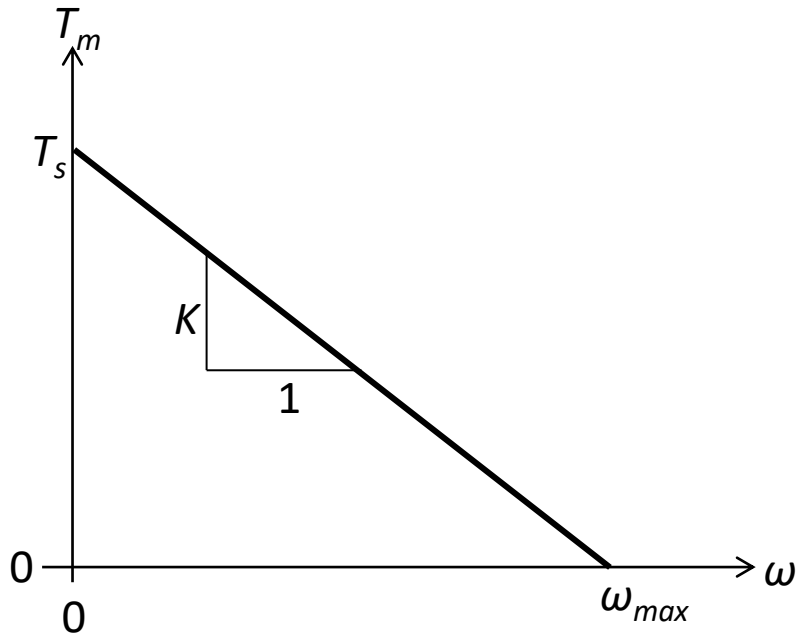


- T_m : Motor Output Torque
- ω : Motor Output Speed
- V_s : Supply voltage
- i_m : Motor Winding Current
- R : Winding Resistance
- L : Winding Inductance
- V_b : Back EMF
- V_R : Voltage across resistor
- V_L : Voltage across inductor
- K_t : Motor's torque constant
- K_b : Motor's back EMF constant
- T_s : Stall torque
- K : Torque-speed gradient

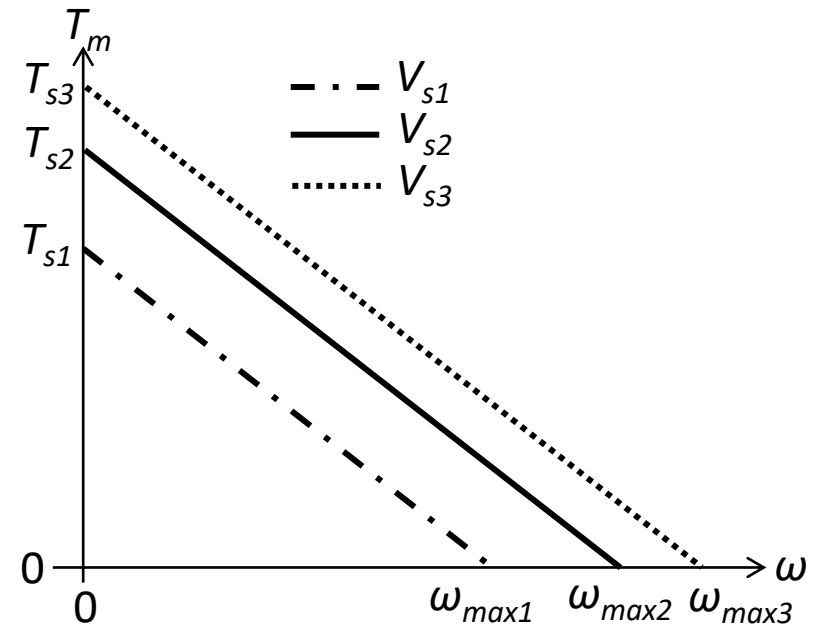
Note that $K_t = K_b$ (if expressed in N-m/A and V/(rad/s), respectively)



Speed-Torque Curve of PMDC Motor



Speed-torque curve
(for a given V_s)



Speed-torque curves
for various V_s values ($V_{s1} < V_{s2} < V_{s3}$)

- **Speed-torque curve** describes relationship between motor speed and torque at
- **Stall Torque** (T_s):; it occurs when the speed of the motor is zero (i.e., motor is stalled).
- (Theoretical) **No-load speed** (ω_{max}): (in theory); it occurs when motor has no load applied to it (i.e., motor is free running).
- T_s and ω_{max} both depend on supply voltage (V_s); they both vary in direct proportion to V_s .
- K does not depend on V_s ; it depends on the motor windings. For a given motor K is fixed.



Exercise 1

For the highlighted motor, calculate T_s (stall torque) and i_s (starting/stall current) from the provided parameters. How do your calculated values compare to those reported in the table?

Assume that the motor operates at the nominal voltage

		118693	118694	118695	118696	118697	118698	118699	118700	118701
Motor Data										
Values at nominal voltage										
1 Nominal voltage	V	1.8	2.4	3	3.2	4.5	4.8	7.2	9	12
2 No load speed	rpm	4990	6360	6890	6270	6740	5700	6890	6740	7130
3 No load current	mA	23.5	25.4	23	18.6	14.8	10.8	9.57	7.4	6.05
4 Nominal speed	rpm	4320	5510	5820	4930	5050	3630	4810	4630	5030
5 Nominal torque (max. continuous torque)	mNm	2.39	2.5	2.89	3.41	4.48	5.61	5.54	5.48	5.48
6 Nominal current (max. continuous current)	A	0.72	0.72	0.72	0.72	0.72	0.711	0.566	0.438	0.348
7 Stall torque	mNm	15.5	16.9	17.3	15.2	17.4	15.2	18.1	17.4	18.6
8 Starting current	A	4.53	4.71	4.19	3.13	2.74	1.9	1.82	1.37	1.16
9 Max. efficiency	%	86	86	86	85	86	86	86	86	86
Characteristics										
10 Terminal resistance	Ω	0.397	0.51	0.715	1.02	1.64	2.53	3.95	6.56	10.3
11 Terminal inductance	mH	0.0207	0.0227	0.0302	0.0415	0.0711	0.113	0.174	0.284	0.452
12 Torque constant	mNm/A	3.43	3.58	4.13	4.84	6.34	7.99	9.92	12.7	16
13 Speed constant	rpm/V	2790	2660	2310	1970	1510	1190	962	753	597
14 Speed / torque gradient	rpm/mNm	323	379	400	415	391	378	383	389	386
15 Mechanical time constant	ms	5.84	5.71	5.56	5.46	5.36	5.31	5.29	5.29	5.27
16 Rotor inertia	gcm ²	1.73	1.44	1.33	1.26	1.31	1.34	1.32	1.3	1.3



Exercise 1: Solution

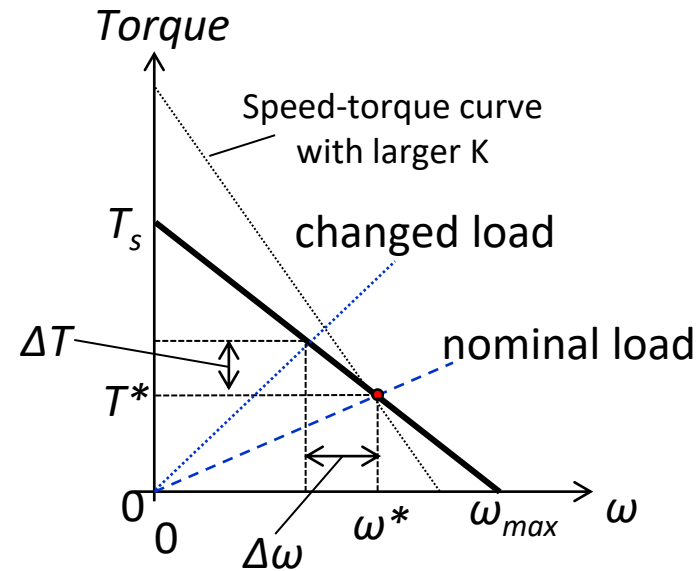
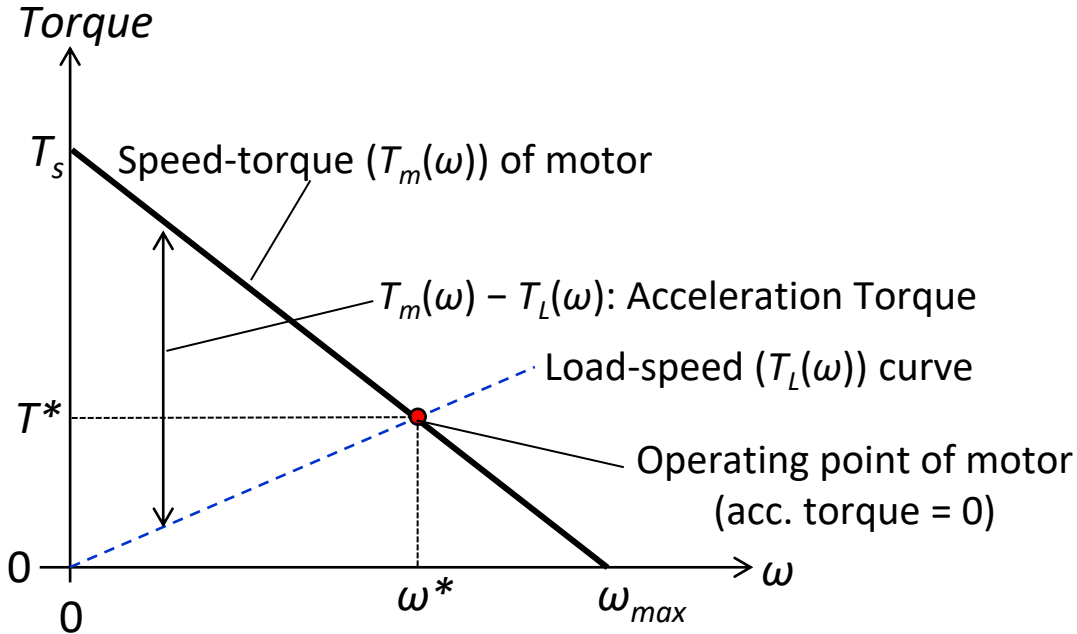
For the highlighted motor, calculate T_s (stall torque) and i_s (starting/stall current) from the provided parameters. How do your calculated values compare to those provided in the table?

Assume that the motor operates at the nominal supply voltage

Motor Data			118701
Values at nominal voltage			
1	Nominal voltage	V	12
2	No load speed	rpm	7130
3	No load current	mA	6.05
4	Nominal speed	rpm	5030
5	Nominal torque (max. continuous torque)	mNm	5.48
6	Nominal current (max. continuous current)	A	0.348
7	Stall torque	mNm	18.6
8	Starting current	A	1.16
9	Max. efficiency	%	86
Characteristics			
10	Terminal resistance	Ω	10.3
11	Terminal inductance	mH	0.452
12	Torque constant	mNm/A	16
13	Speed constant	rpm/V	597
14	Speed / torque gradient	rpm/mNm	386
15	Mechanical time constant	ms	5.27
16	Rotor inertia	gcm ²	1.3



Effect of Load on Motor



- When load is applied, starting from rest motor accelerates in proportion to (which in general varies with speed)
- Motor reaches its steady-statewhen $T_m = T_L = T^*$ (i.e., acceleration torque = 0); the corresponding speed is ω^*
- refers to the ability to maintain its operating speed as load is varied. It is given in percentage by $SR = 100 * (\omega_{max} - \omega^*) / \omega^*$.
- Note that as K increases, speed regulation improves (i.e., SR decreases) and $\Delta\omega / \Delta T$ decreases, meaning that smaller speed fluctuation ($\Delta\omega$) is experienced for a given load fluctuation (ΔT)

Exercise 2

Calculate the operating speed if the motor highlighted in the spec sheet below is used to drive a load with torque characteristics given by

$$T_L = A + B\omega, \text{ where } A = 2 \text{ mN-m and } B = 0.001 \text{ mN-m/rpm}$$

Assume that the motor operates at nominal voltage

		118693	118694	118695	118696	118697	118698	118699	118700	118701
Motor Data										
Values at nominal voltage										
1 Nominal voltage	V	1.8	2.4	3	3.2	4.5	4.8	7.2	9	12
2 No load speed	rpm	4990	6360	6890	6270	6740	5700	6890	6740	7130
3 No load current	mA	23.5	25.4	23	18.6	14.8	10.8	9.57	7.4	6.05
4 Nominal speed	rpm	4320	5510	5820	4930	5050	3630	4810	4630	5030
5 Nominal torque (max. continuous torque)	mNm	2.39	2.5	2.89	3.41	4.48	5.61	5.54	5.48	5.48
6 Nominal current (max. continuous current)	A	0.72	0.72	0.72	0.72	0.72	0.711	0.566	0.438	0.348
7 Stall torque	mNm	15.5	16.9	17.3	15.2	17.4	15.2	18.1	17.4	18.6
8 Starting current	A	4.53	4.71	4.19	3.13	2.74	1.9	1.82	1.37	1.16
9 Max. efficiency	%	86	86	86	85	86	86	86	86	86
Characteristics										
10 Terminal resistance	Ω	0.397	0.51	0.715	1.02	1.64	2.53	3.95	6.56	10.3
11 Terminal inductance	mH	0.0207	0.0227	0.0302	0.0415	0.0711	0.113	0.174	0.284	0.452
12 Torque constant	mNm/A	3.43	3.58	4.13	4.84	6.34	7.99	9.92	12.7	16
13 Speed constant	rpm/V	2790	2660	2310	1970	1510	1190	962	753	597
14 Speed / torque gradient	rpm/mNm	323	379	400	415	391	378	383	389	386
15 Mechanical time constant	ms	5.84	5.71	5.56	5.46	5.36	5.31	5.29	5.29	5.27
16 Rotor inertia	gcm ²	1.73	1.44	1.33	1.26	1.31	1.34	1.32	1.3	1.3



Exercise 2: Solution

Calculate the operating speed if the motor highlighted in the spec sheet below is used to drive a load with torque characteristics given by

$$T_L = A + B\omega, \text{ where } A = 2 \text{ mN-m and } B = 0.001 \text{ mN-m/rpm}$$

Assume that the motor operates at nominal voltage

Motor Data			118701
Values at nominal voltage			
1	Nominal voltage	V	12
2	No load speed	rpm	7130
3	No load current	mA	6.05
4	Nominal speed	rpm	5030
5	Nominal torque (max. continuous torque)	mNm	5.48
6	Nominal current (max. continuous current)	A	0.348
7	Stall torque	mNm	18.6
8	Starting current	A	1.16
9	Max. efficiency	%	86
Characteristics			
10	Terminal resistance	Ω	10.3
11	Terminal inductance	mH	0.452
12	Torque constant	mNm/A	16
13	Speed constant	rpm/V	597
14	Speed / torque gradient	rpm/mNm	386
15	Mechanical time constant	ms	5.27
16	Rotor inertia	gcm ²	1.3



Exercise 3

What supply voltage is needed to operate the highlighted motor at 3000 rpm if it drives the same load as in Exercise 2 (with torque characteristics given by

$$T_L = A + B\omega, \text{ where } A = 2 \text{ mN-m and } B = 0.001 \text{ mN-m/rpm})$$

		118693	118694	118695	118696	118697	118698	118699	118700	118701
Motor Data										
Values at nominal voltage										
1 Nominal voltage	V	1.8	2.4	3	3.2	4.5	4.8	7.2	9	12
2 No load speed	rpm	4990	6360	6890	6270	6740	5700	6890	6740	7130
3 No load current	mA	23.5	25.4	23	18.6	14.8	10.8	9.57	7.4	6.05
4 Nominal speed	rpm	4320	5510	5820	4930	5050	3630	4810	4630	5030
5 Nominal torque (max. continuous torque)	mNm	2.39	2.5	2.89	3.41	4.48	5.61	5.54	5.48	5.48
6 Nominal current (max. continuous current)	A	0.72	0.72	0.72	0.72	0.72	0.711	0.566	0.438	0.348
7 Stall torque	mNm	15.5	16.9	17.3	15.2	17.4	15.2	18.1	17.4	18.6
8 Starting current	A	4.53	4.71	4.19	3.13	2.74	1.9	1.82	1.37	1.16
9 Max. efficiency	%	86	86	86	85	86	86	86	86	86
Characteristics										
10 Terminal resistance	Ω	0.397	0.51	0.715	1.02	1.64	2.53	3.95	6.56	10.3
11 Terminal inductance	mH	0.0207	0.0227	0.0302	0.0415	0.0711	0.113	0.174	0.284	0.452
12 Torque constant	mNm/A	3.43	3.58	4.13	4.84	6.34	7.99	9.92	12.7	16
13 Speed constant	rpm/V	2790	2660	2310	1970	1510	1190	962	753	597
14 Speed / torque gradient	rpm/mNm	323	379	400	415	391	378	383	389	386
15 Mechanical time constant	ms	5.84	5.71	5.56	5.46	5.36	5.31	5.29	5.29	5.27
16 Rotor inertia	gcm ²	1.73	1.44	1.33	1.26	1.31	1.34	1.32	1.3	1.3



Exercise 3: Solution

What supply voltage is needed to operate the highlighted motor at 3000 rpm if it drives the same load as in Exercise 2 (with torque characteristics given by

$$T_L = A + B\omega, \text{ where } A = 2 \text{ mN-m and } B = 0.001 \text{ mN-m/rpm})$$

Motor Data			118701
Values at nominal voltage			
1	Nominal voltage	V	12
2	No load speed	rpm	7130
3	No load current	mA	6.05
4	Nominal speed	rpm	5030
5	Nominal torque (max. continuous torque)	mNm	5.48
6	Nominal current (max. continuous current)	A	0.348
7	Stall torque	mNm	18.6
8	Starting current	A	1.16
9	Max. efficiency	%	86
Characteristics			
10	Terminal resistance	Ω	10.3
11	Terminal inductance	mH	0.452
12	Torque constant	mNm/A	16
13	Speed constant	rpm/V	597
14	Speed / torque gradient	rpm/mNm	386
15	Mechanical time constant	ms	5.27
16	Rotor inertia	gcm ²	1.3

